EXP 1.

Select any system and thoroughly examine its system specifications to identify and document the different software defects present.

FEATURES TO BE TESTED:

1. Validity of the ATM Card.
2. Withdrawal transaction flow of the ATM.
3. Authentication of the user.
4. Dispensing cash from the account.
5. Account balance verification.
6. Withdrawal limit per transaction.
7. Daily withdraw limit.
8. Valid money denominations.
9. Receipt printing.
10. PIN change functionality.

Bug ID	Bug name
ATM 001	Invalid Card.
ATM 002	Invalid PIN
ATM 003	Invalid Account Type.
ATM 004	Insufficient Balance.
ATM 005	Transaction Limit Exceeded.
ATM 006	Daily withdrawal Limit Exceeded.
ATM 007	Invalid Money Denominations.
ATM 008	Receipt Not Printed.
ATM 009	PIN Change Mismatch.

Bug reports.

Bug ID: ATM001

* Bug Description: Invalid Card.
* Steps to Reproduce:
  Insert an invalid or unrecognized card into the ATM.
* Expected Result:
  The ATM should display an "Invalid Card" message and ~~deny access~~. reject the card.
* Actual Result: The ATM displays the welcome screen if the defect occurs.
* Status: Pass/Fail.

Bug ID: ATM002.

* Bug Description: Invalid PIN.
* Steps to Reproduce:
  * Insert a valid ATM card.
  * Enter an incorrect PIN.
  * Submit the PIN.
* Expected Result: ATM should display "Invalid PIN". and deny access.
* Actual Result: ATM displays the menu screen if the defects occurs.

Bug ID: ATM003.

Bug Description: Invalid Account Type.

Steps to Reproduce:

* Insert a valid ATM card and enter the correct PIN.
* Select the withdrawal option from the main menu.
* Select an unsupported or invalid account type.

Expected Result:
The ATM should reject the invalid account type and display an appropriate error message.

Actual Result:
The ATM displays the amount-entry screen if the defect occurs.

Status: Pass / Fail.

Bug ID: ATM 004

* Bug Description: Insufficient Balance.
* Steps to Reproduce:
  * Insert a valid ATM card and enter the correct PIN.
  * Select the withdrawal option
  * Select a valid account type.
  * Enter an amount greater than the available balance.
  * Confirm the transaction.

* Expected Result:
ATM should display an "Insufficient Balance" msg. and reject the transaction.

* Actual Result:
ATM displays the cash collection screen & deducts the amount if the defect occurs.

* Status: Pass/Fail.

BUG ID: ATM 005.

* Bug Description: Withdrawal limit per Transaction.
* Steps to Reproduce:
  * Insert a valid ATM card and enter the correct PIN.
  * Select the withdrawal option.
  * Select a valid account type.
  * Enter an amount within the permitted transaction limit
  * Confirm the transaction.

* Expected Result: ATM should dispense cash and provide a receipt if sufficient balance and cash are available.

* Actual Result: ATM displays a "Transaction Limit Exceeded" msg if the defect occurs.

* Status: Pass / Fail

Bug ID: ATM 006.

* Bug Description: Daily Withdrawal Limit.

* Steps to reproduce:
  * Insert a valid ATM card.
  * Enter the correct PIN.
  * Select the withdrawal option.
  * Enter amt that causes the total daily withdrawal to exceed the configured limit.
  * Confirm the transaction.

* Expected Result:
  ATM should reject a transaction that exceeds the daily withdrawal limit and display an appropriate msg.

* Actual Result: ATM dispenses cash despite the daily limit being exceeded if the defect occurs.

* Status: Pass / Fail.

Bug ID: ATM 007.

Bug Description: Amount enter denominations.

Steps to reproduce: Keep a valid card in ATM.
Enter the authorized PIN.
Enter amount which should be in multiples of 100.
Cash dispenses screen is displayed.

Expected o/p: Collect the amount screen is displayed.

Actual o/p: Amount enter not in required denominations.

2. Write a case study to Identify and analyze the requirements of ARMS Portals, and prepare positive test scenarios as well as test cases for the application.

FEATURES TO BE TESTED:

* Login functionality.
* Profile viewing.
* Attendance details.
* Examination results.
* Academic record management.
* Logout functionality.

POSITIVE TEST SCENARIOS:

Test Scenario ID.	Test Scenario.
ARMS_P01	Login with valid credentials.
ARMS_P02	View student profile.
ARMS_P03	View attendance details.
ARMS_P04	View published examination results.
ARMS_P05	Update academic records as authorized staff.
ARMS_P06	Logout successfully.

Test Case ID: ARMS_P01.

* Test case Name: Login with valid credentials.
* Precondition: The user has a registered account.
* steps to Execute:
  * Open the ARMS Portal.
  * Enter a valid username.
  * Enter correct password.
  * Click the Login button.
* Expected op: The user should be authenticated & redirected to the dashboard.
* Actual Result: To be recorded after execution.

Test Case ID : ARMS_P02.

* Test Case Name : View Student profile
* Precondition : The student is logged in.
* Steps to Execute :
    * Open the dashboard.
    * Click the profile option.
    * View the displayed personal details.
* Expected Result : system should display correct name, reg. no, dep & other details.
* Actual Result : To be recorded after execution.
* Status : Pass/Fail.

Test Case ID : ARMP03.

* Test Case Name : View attendance details.
* Steps to Execute :
    * Navigate to the Attendance section.
    * Select the required semester or subject. if applicable
    * View the attendance details.
* Expected o/p : System should display the correct attendance info.
* Actual o/p : Correct info was seen.
* Status : Pass/Fail.

Test Case ID : ARMS_P04.

* Test Case Name : View published examination results
* Steps to Execute.
    * Login to the portal
    * Open the Examination result section.
    * Select the relevant semester.
    * View result
* Expected o/p : The system should display correct mark & status
* Actual o/p : Correct subject-wise marks and result status was seen.
* Status : Pass/Fail.

Test Case ID : ARMS_P05.

* Test Case Name : Update academic records
* Steps to Execute :
  * Open the student record management section.
  * Search for the required student.
  * Enter valid and authorized academic info.
  * Click save.
* Expected O/P : System should save the changes & display it.
* Actual O/P : The updated info was displayed on the system.
* Status : Pass/Fail.

Test Case ID : ARMS_P06.

* Test Case Name : Logout from the portal.
* Steps to Execute : The user is logged in.
  * Click the Logout option.
  * Attempted to access the protected dashboard using previous session.
* Expected O/P : User should be logged out.
* Actual O/P : User was logged out & dashboard should no longer be accessible.
* Status : Pass/Fail.

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3) ARMS PORTAL - NEGATIVE SCENARIOS & TEST CASES.

Negative Test Cases :

Test Case ID : NT01.

* Test Case : Login with incorrect password.
* Precondn : User must have registered account.
* Steps to Execute :
  * Open ARMS portal.
  * Enter valid username.
  * Enter incorrect password.
  * Click login.
* Test Data : Valid username & Incorrect Password.
* Expected Result : Login should fail, appropriate error msg.
* Status : Pass/Fail.

Test Case ID : NT02.

* Test Case : Login with unregistered username.
* Precondn : ARMS portal login page must available.
* Steps to Execute :
    * Open the portal.
    * Enter an unregistered username.
    * Enter a password.
    * Click Login.
* Test Data : Unregistered username
* Expected Result : System should reject login attempt.
* Status : Pass/Fail.

Test Case ID : NT03.

* Test Cases : Submit login from with empty fields.
* Precondn : Login page must displayed.
* Steps to Execute :
    * Open the portal.
    * Leave username, password fields Empty.
    * Click Login.
* Test Data : Empty username, password.
* Expected Result : System should display req-field valiation & login,
* Status : Pass/Fail.

Test Case ID : NT04.

* Test Case : Enter invalid register number
* Precondn : Submit registration (or) search form.
* Steps to Execute :
    * Open student details form.
    * Enter invalid reg no.
    * Click submit or search.

* Test Data : @@@###
* Expected Result : System should reject invalid reg no, display msg.
* Status : Pass/Fail.

Test Case ID : NTOS

* Test Case : Access another students academic records.
* Precondition : Student must be logged in.
* Steps to Execute :
    * Login using student account.
    * Attempt access another students record by modifying Student ID in request or URL.
    * Submit the request.
* Test Data : Another students ID.
* Expected Result : System should deny access, protect students info
* Status : Pass/Fail.

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4) E-commerce Application - Positive test scenario & test case.

Positive Test Cases :

Test Case ID : ETC01

* Test Case : Registers with valid details.
* Precondition : Reg page must be available.
* Steps to Execute :
    * Open E-commerce application.
    * Click Register.
    * Enter valid name, email & password.
    * Click submit.
* Test Data : Valid name, unique-email, valid password.
* Expected Result : account should be created successfully
* Status : Pass/Fail.

Test Case ID: ETC02.

*Test Case: Login with valid credentials.

*Precondn: User must have a registered account.

*Steps to Execute:

* open login page.
* Enter registered email & correct password.
* click Login.

*Test Data: Valid email & password

*Expected Result: User should log successfully, home page display

*Status: Pass/Fail.

Test Case ID: ETC03.

*Test Case: Search for existing product.

*Precondn: Products must available in location.

*steps to Execute:

* open Application.
* Enter product name in search bar.
* click Search.

*Test Data Laptop.

*Expected Result: Relevant products should displayed.

*Status: Pass/Fail.

Test Case ID: ETC04.

*Test Case: View product details.

*Precondn: Select required product.

*Steps to Execute:

* Search for product.
* Select required product.
* Open details page.

*Test Data: An existing product.

* Expected Result: Correct product name, descrip. price be display
* Status: Pass/Fail.

Test Case ID: ETC05.

* Test Case: Add product to existing shopping cart.
* Precondn: Product must be available in stock.
* Steps to Execute:
    * Open product page.
    * Click Add to cart.
    * Open shopping cart.
* Test Data: Available product, Quantity 1.
* Expected Result: Selected product should appear in cart
* Status: Pass/Fail.

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5) E-commerce Application - Negative Test Scenario & Test Cases.

Negative Test Scenario:

Test Case ID: NTC01.

* Test Case: Register with invalid email id.
* Precondition: registration page must be available.
* Steps to Execute:
    1. Open registration Page.
    2. Enter name & invalid email address.
    3. Enter valid Password.
    4. Click submit.
* Test Data: Email abc123
* Expected Result: Application should reject email & display msg.
* Status: Pass/Fail.

Test case ID : NTC02.

* Test-case : Login with incorrect password.
* Precondn : User must have registered account.
* Steps to reproduce :
    * Open login page
    * Enter registered email.
    * Enter incorrect password.
    * Click Login.
* Test Data : Registered email & incorrect password.
* Expected Result : Login should fail.
* Status : Pass/Fail.

Test case ID : NTC03.

* Test case : Submit Registration with empty fields.
* Precondn : Registration page must be displayed.
* Steps to Execute :
    * Open reg page.
    * Leave req. fields empty.
    * Click submit.
* Test Data : Empty Name, email & password.
* Expected Result : Application shld display req field msg
* Status : Pass/Fail.